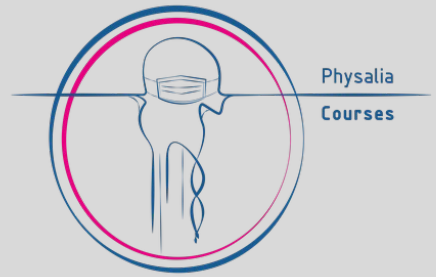
Processing NGS data

Epigenomics Data Analysis

Jacques Serizay

Physalia 2026



NGS processing workflow

- Get .bcl files
- Create fastq files
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- Assay-specific downstream analysis

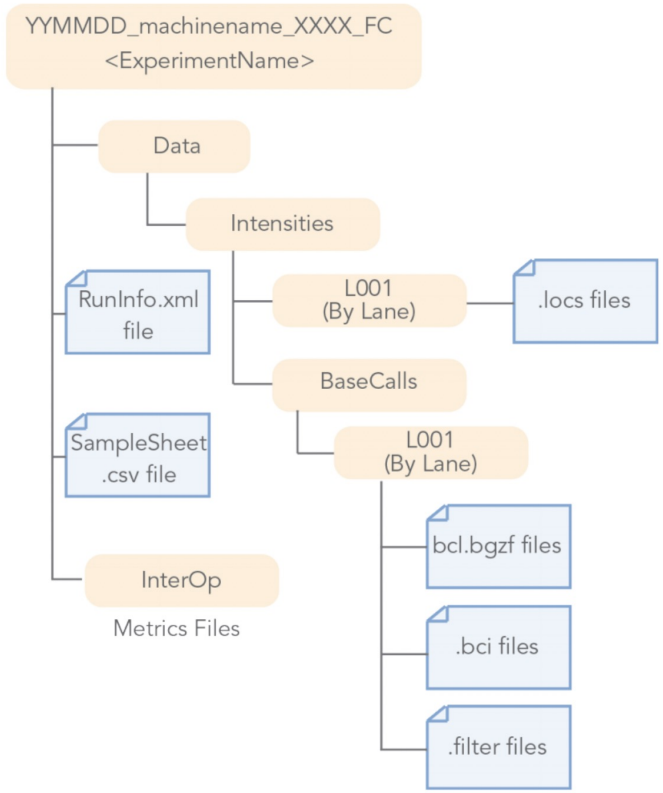
.bcl files

.bcl:

- **Raw data** output of a sequencing run
- **Binary**, non-human-readable file

- Contains the **base calling** and **quality score per cluster, per sequencing lane, per cycle**

- **Huge files**
- **No aggregated sequence per read**



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Fastq files

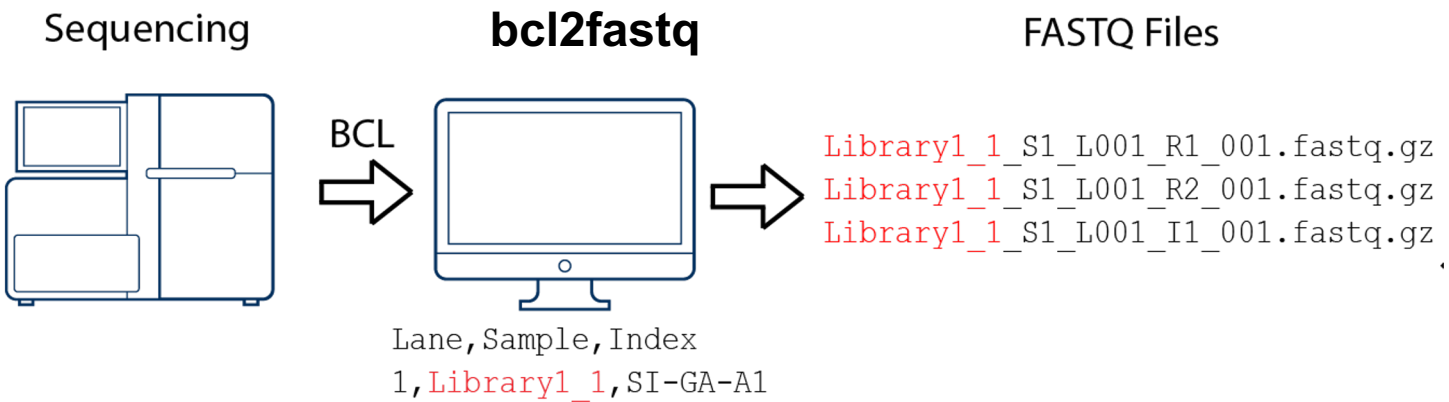
A fastq file contains reads, each read is composed of 4 lines:

1. A sequence identifier with information about the sequencing run
2. The sequence (the base calls; A, C, T, G and N).
3. A separator, which is simply a plus (+) sign.
4. The base call quality scores, using ASCII characters to represent the numerical quality scores.

```
jacquesserizay@LOCAL[12:46:19]:~ $ cat SRR11575369_1.fastq.gz | zcat |  head -n 8
@SRR11575369.1 1/1
ANCAACAGTGGAATTCGTTTTGATGAAAAAAATAAATTTGTTCTTCAAAGCAGAGTGAATGATGCAGTACGAGCTCTTGCTCTTGAAAACCCATCACAACTTTATAATTTAATAATTAGTGAAAAATTAAAAAAATAATATTTCTTATATT
+
F#F:FFFFF:FFFFFFFFFF:F,FF,FF::F,::FFFFFFFFFFF:FF:FF:F,FFF,F:F,FFF,FF:FF:F:FFF:FF:FFFF,:F::FFFF:FF,FF:F:FF,,:F:F,::,F:F:FF,,::,FF:,,F,,,,,:FFFF::F:,,,FF
@SRR11575369.2 2/1
TNGCCAGTCAATACCGCCCTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTGTGTTTTTTTTTTTTTATATAATAAAATTTTTTTTTTTTTTAATAAAATTTTTTTTTTTATTTT
+
F#FFFFFFF::FF:FF::FFFFFFFFFFFFFFFF:FFFFFFFFFFFFF,,FFFFFFFFF:FFFFFF:FFFFFFFFFFF:F,FFFFF,,,:,,::FFFFFFFF,,:,,,F,,,,,,FFFFFF,F,,FF:,F,:,F,FFFFF,,,:,,F::F
```

bcl2fastq

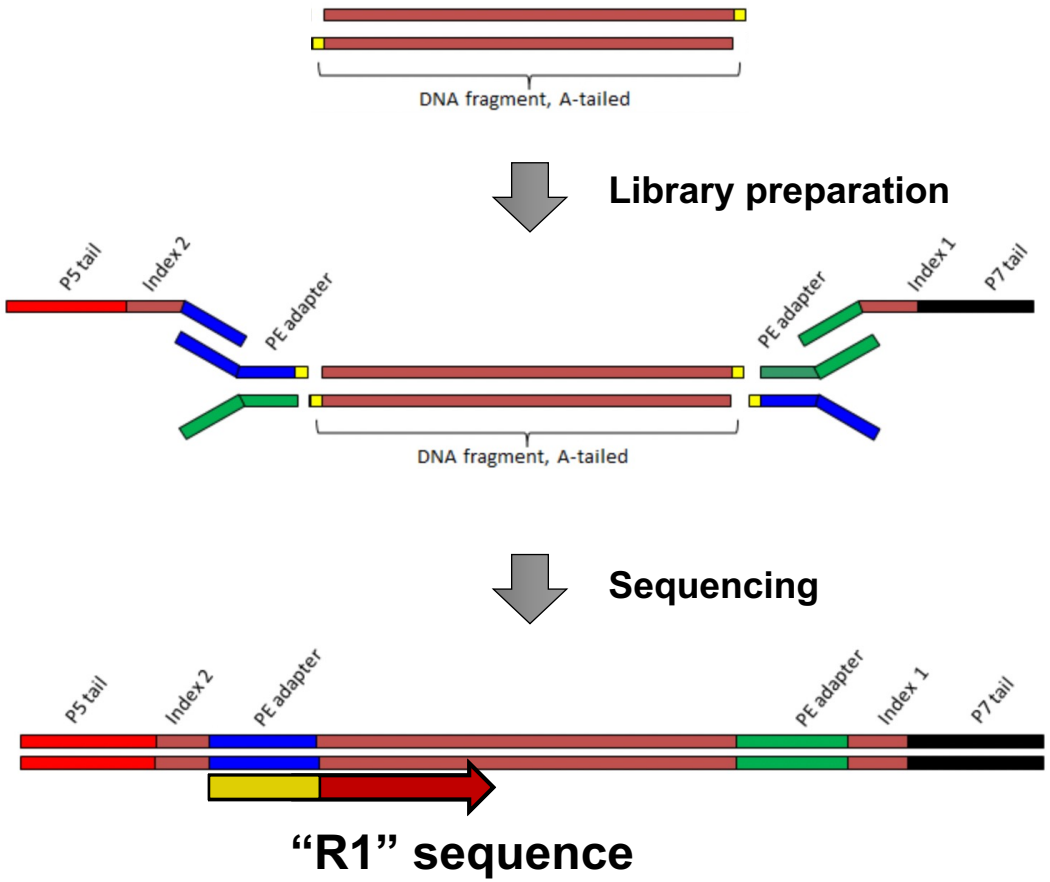
```
bcl2fastq --run-folder-dir <bcl_files_folder> --output-dir <fastq_files_folder>
```



User guide:

https://support.illumina.com/content/dam/illumina-support/documents/documentation/software_documentation/bcl2fastq/bcl2fastq_letterbooklet_15038058brpmi.pdf

Why so many fastq files?



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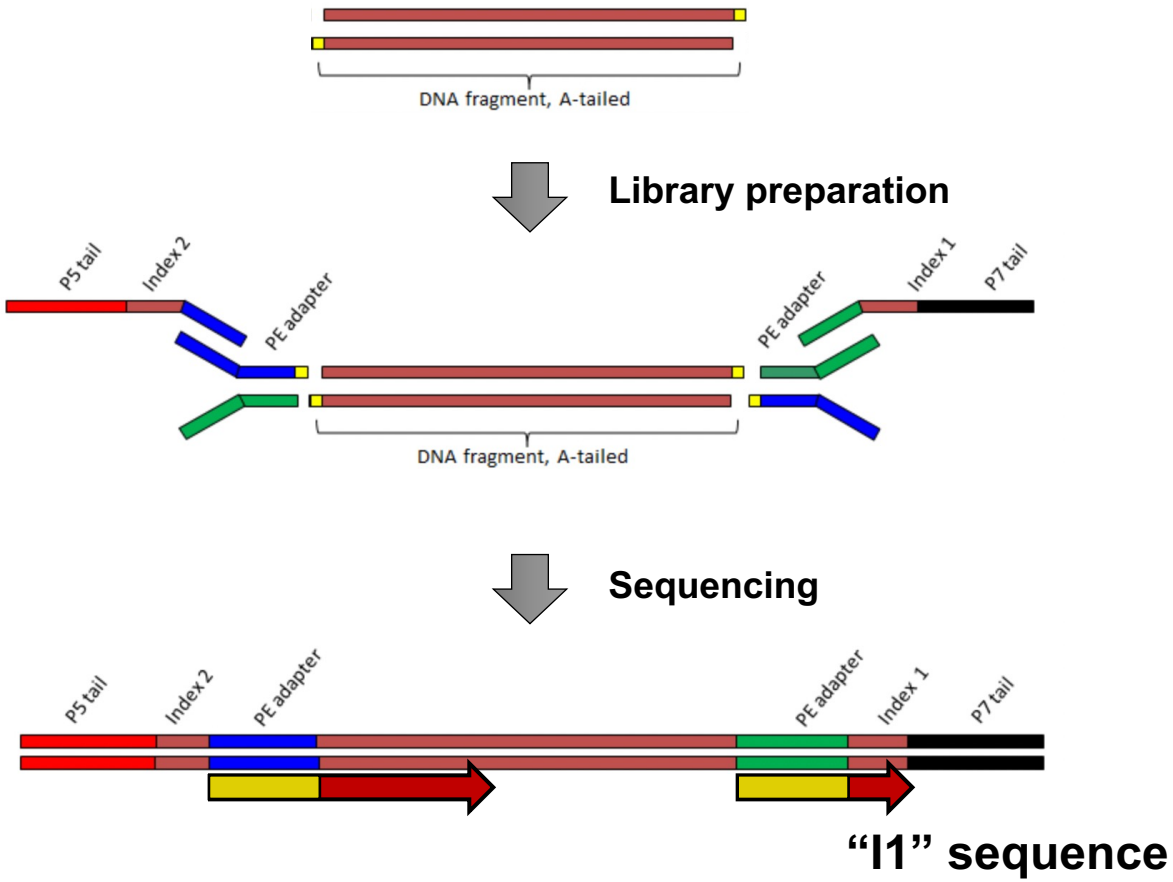

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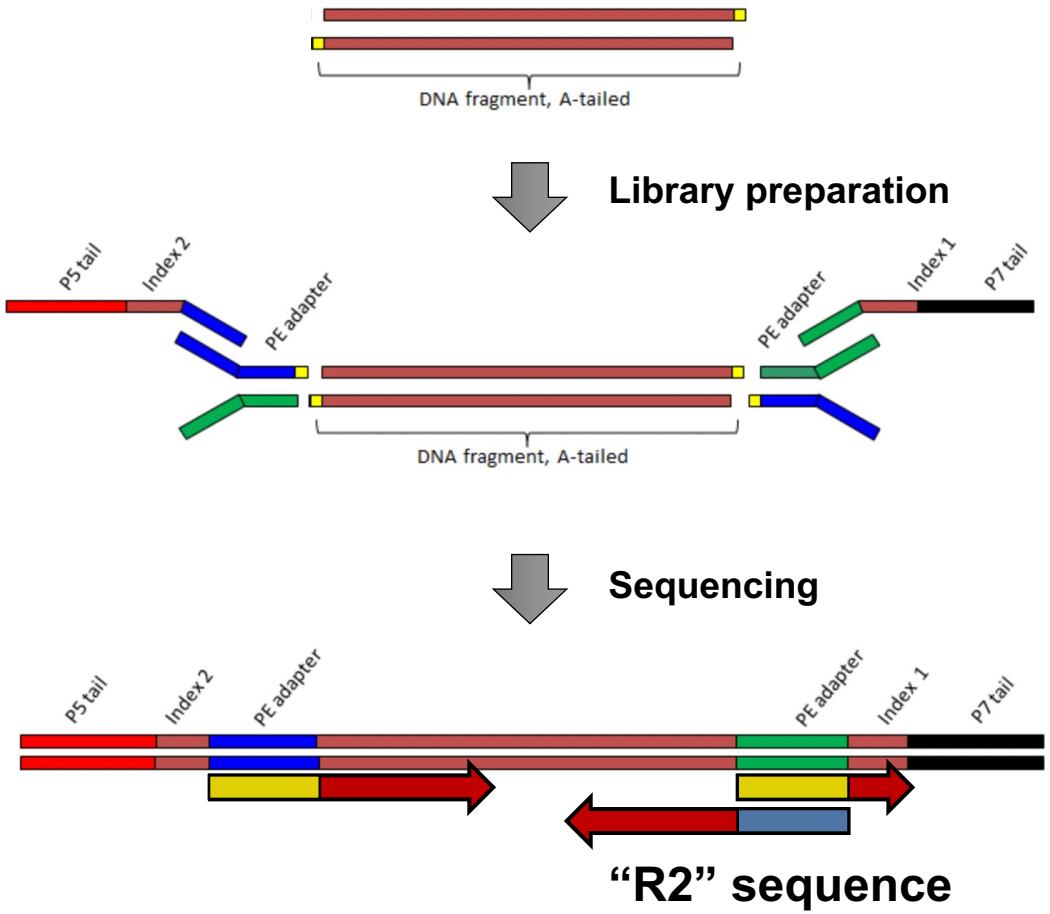

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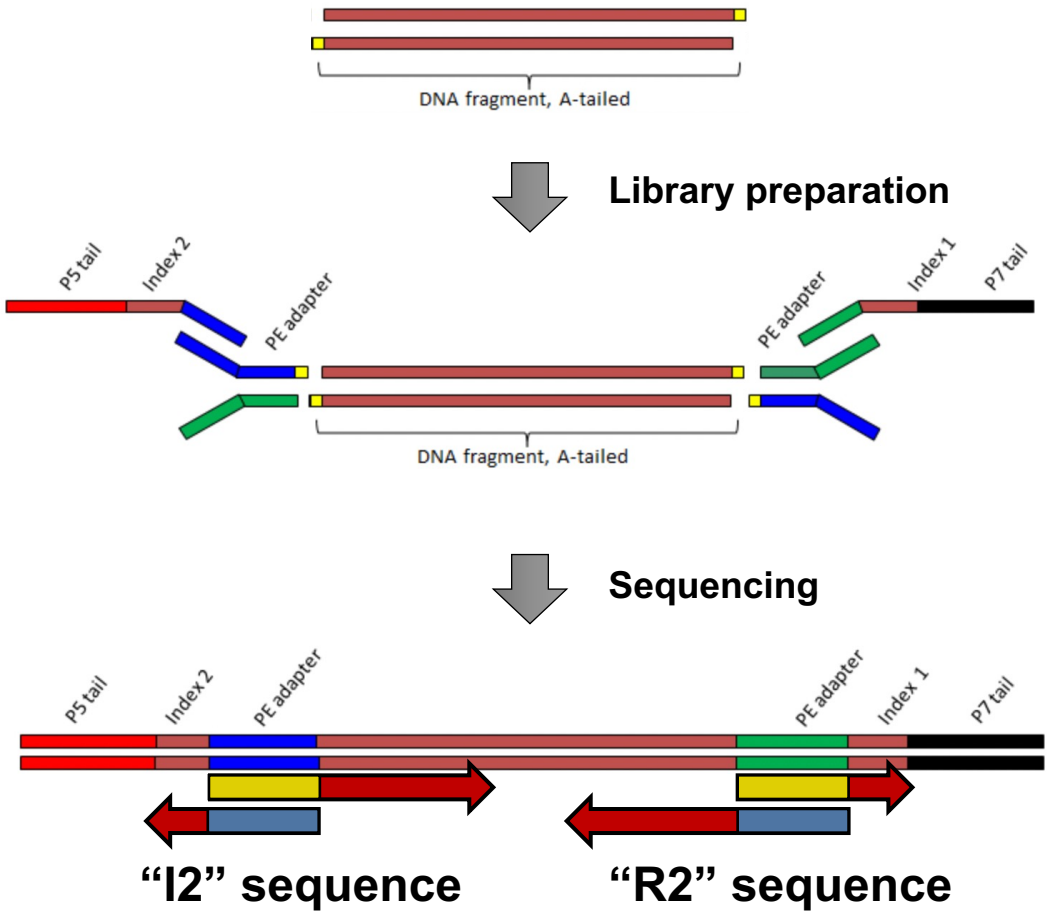

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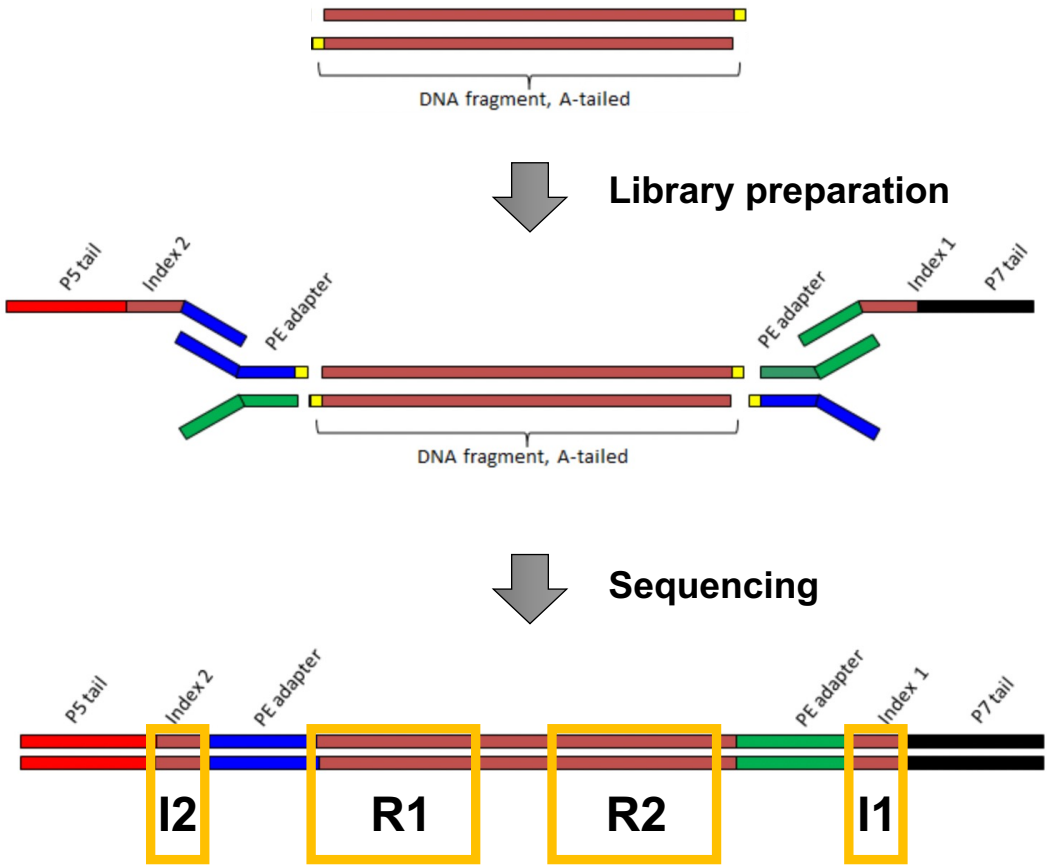

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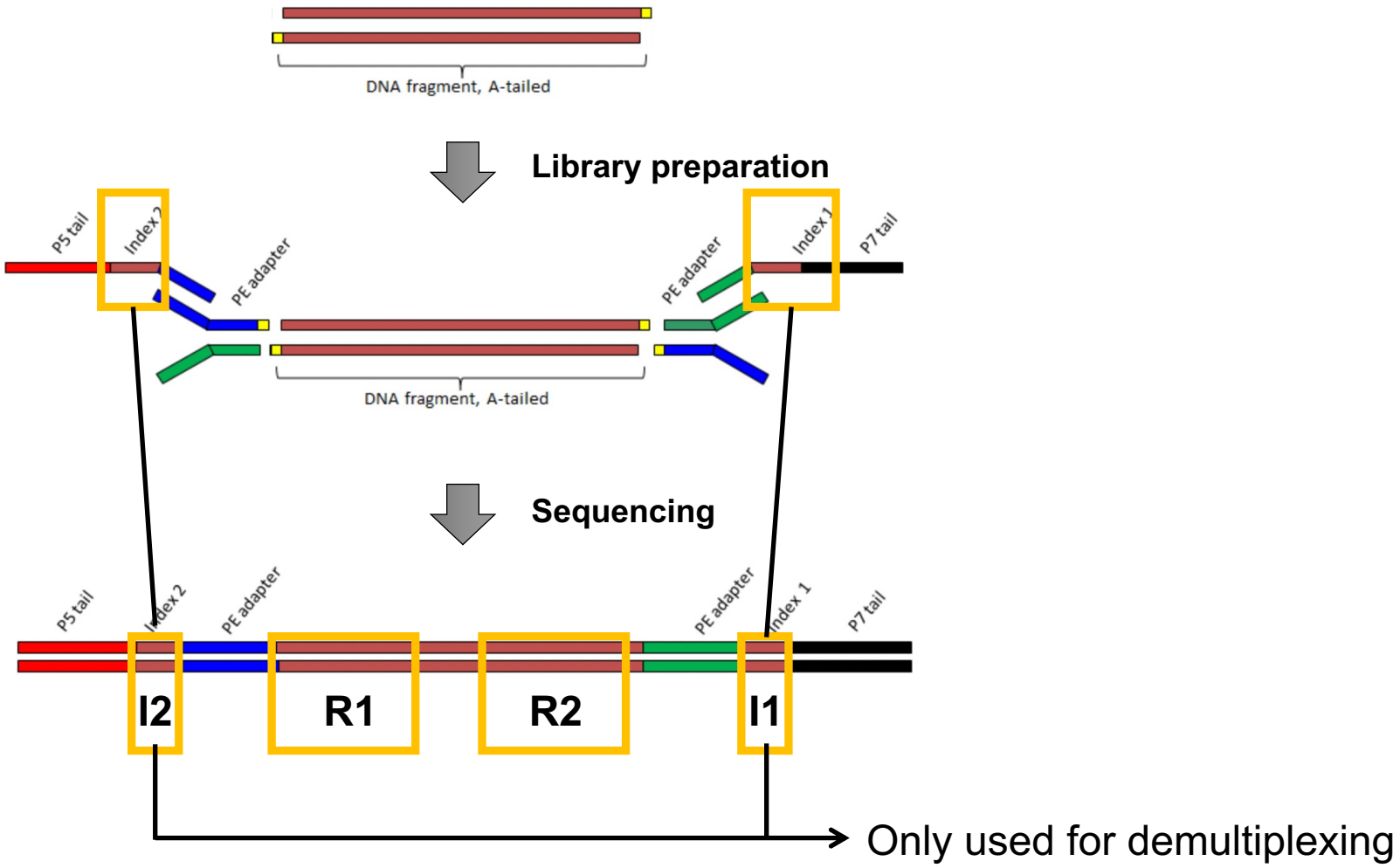

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Or **`bcl2fastq`**

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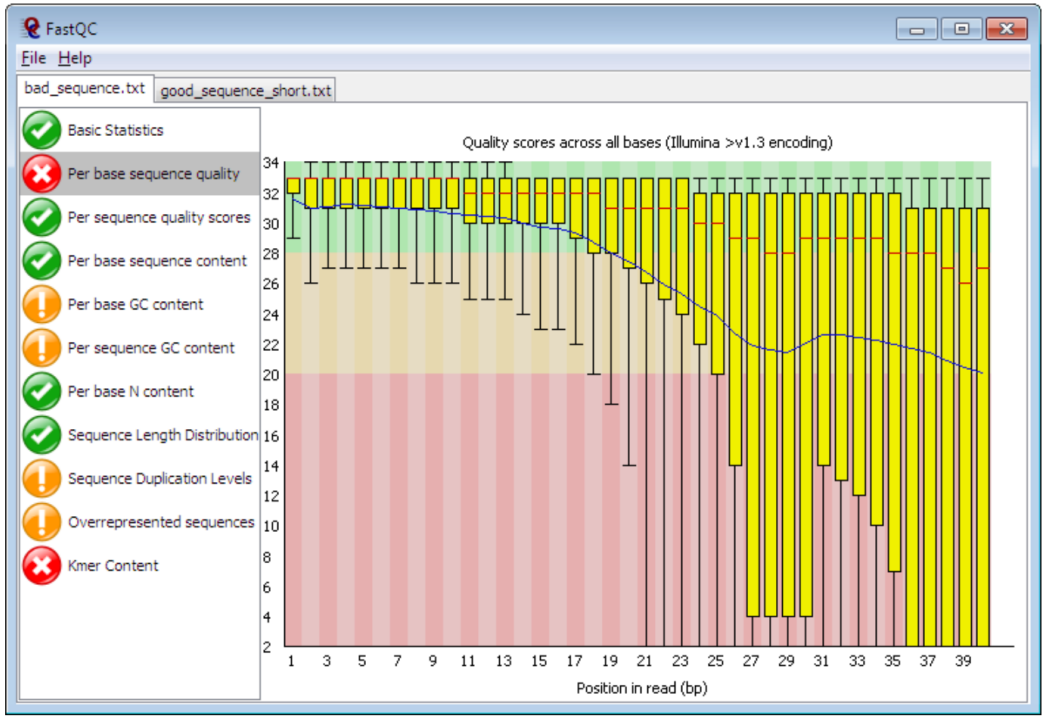
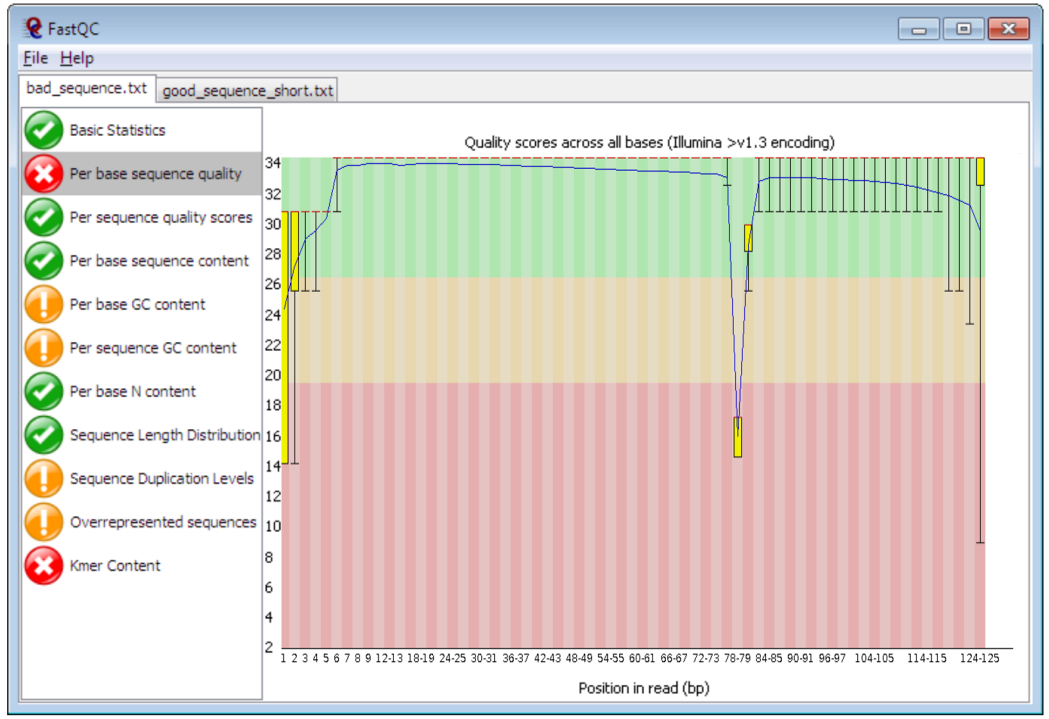
FastQC

FastQC is a program designed to spot potential problems in high througput sequencing datasets. It runs a set of analyses on one or more raw sequence files in fastq or bam format and produces a report which summarises the results.





FastQC will highlight any areas where this library looks unusual and where you should take a closer look. The program is not tied to any specific type of sequencing technique and can be used to look at libraries coming from a large number of different experiment types (Genomic Sequencing, ChIP-Seq, RNA-Seq, BS-Seq etc etc).

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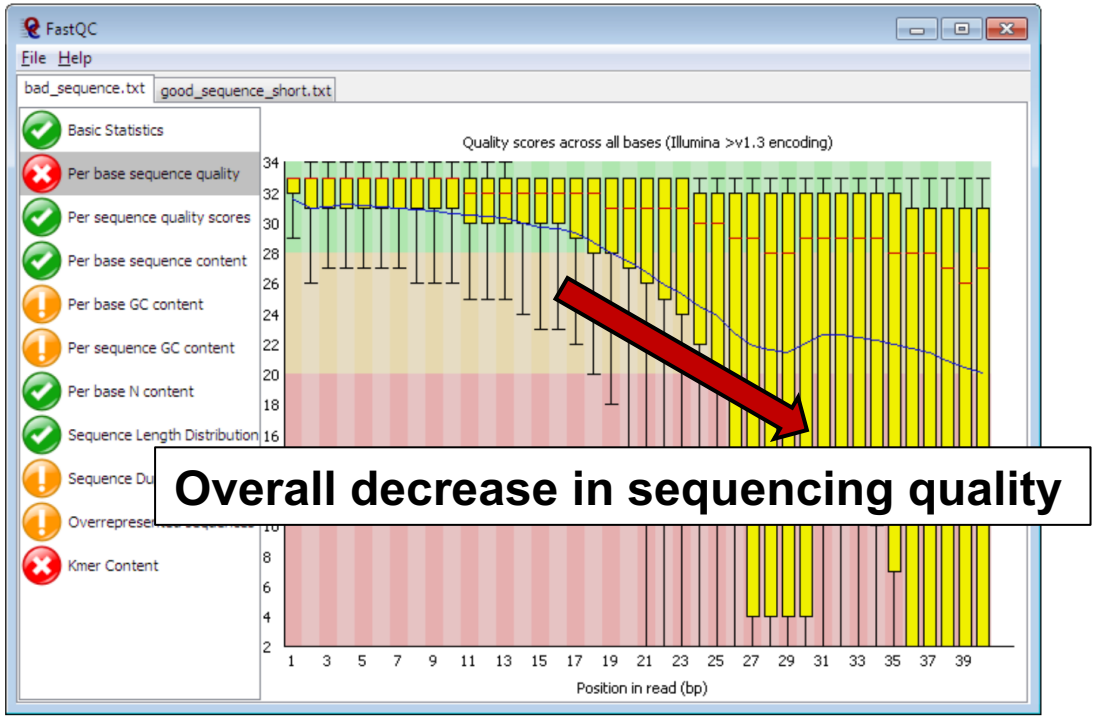

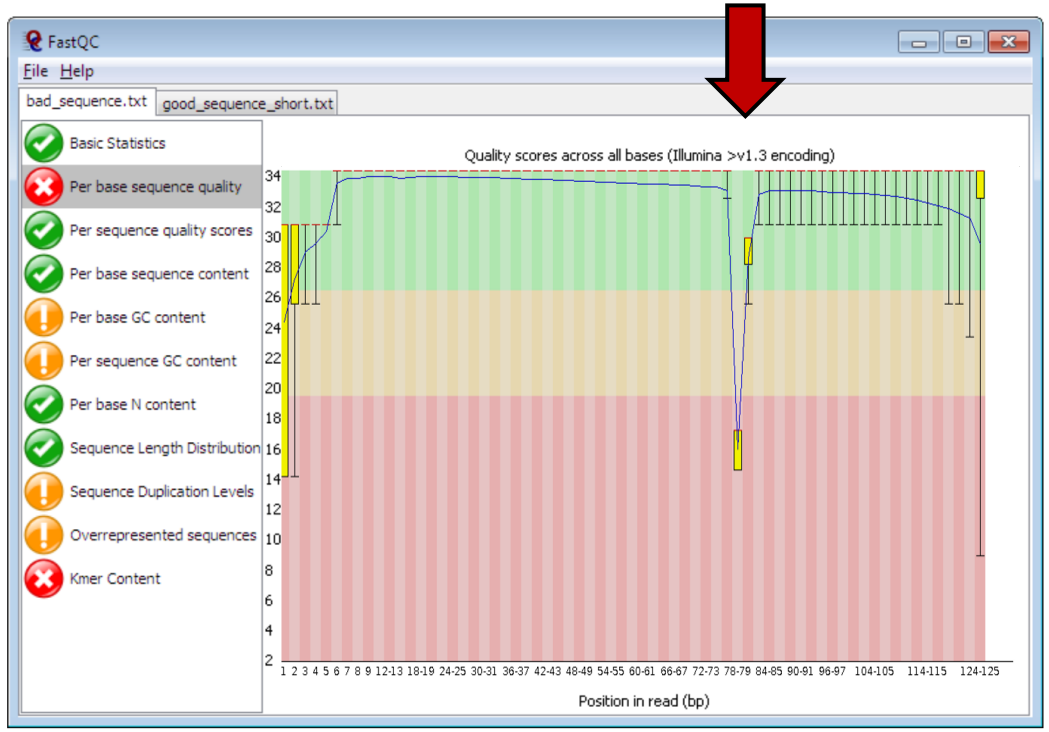

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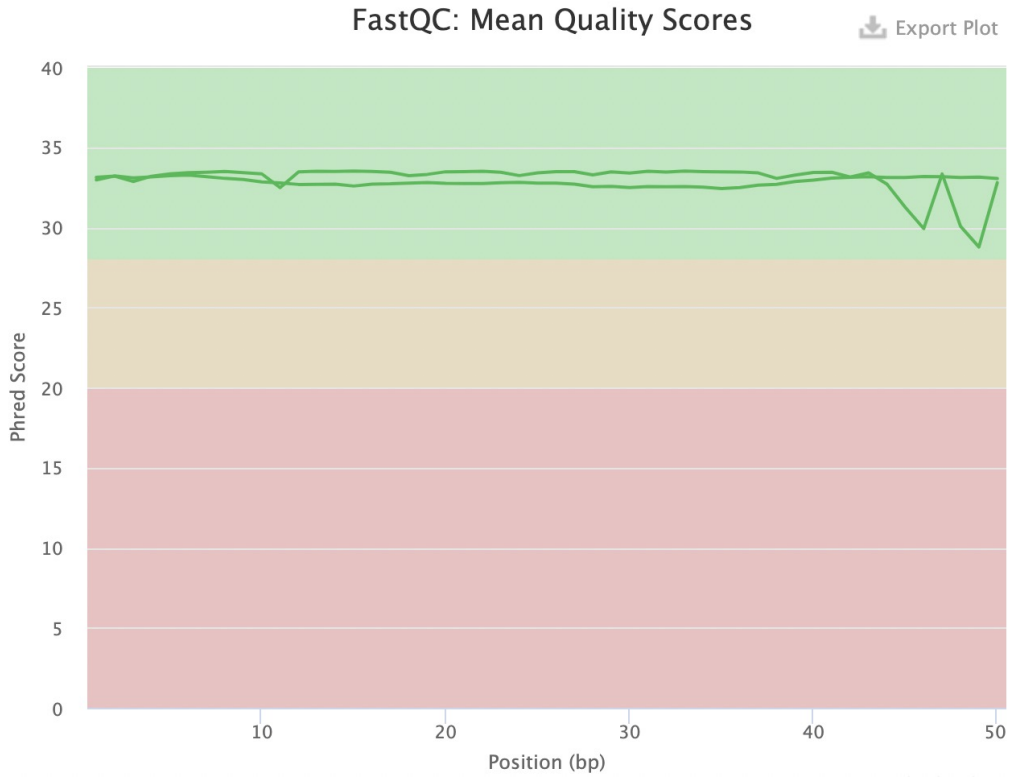
Cutadapt: trim away adapter sequences and low-quality ends



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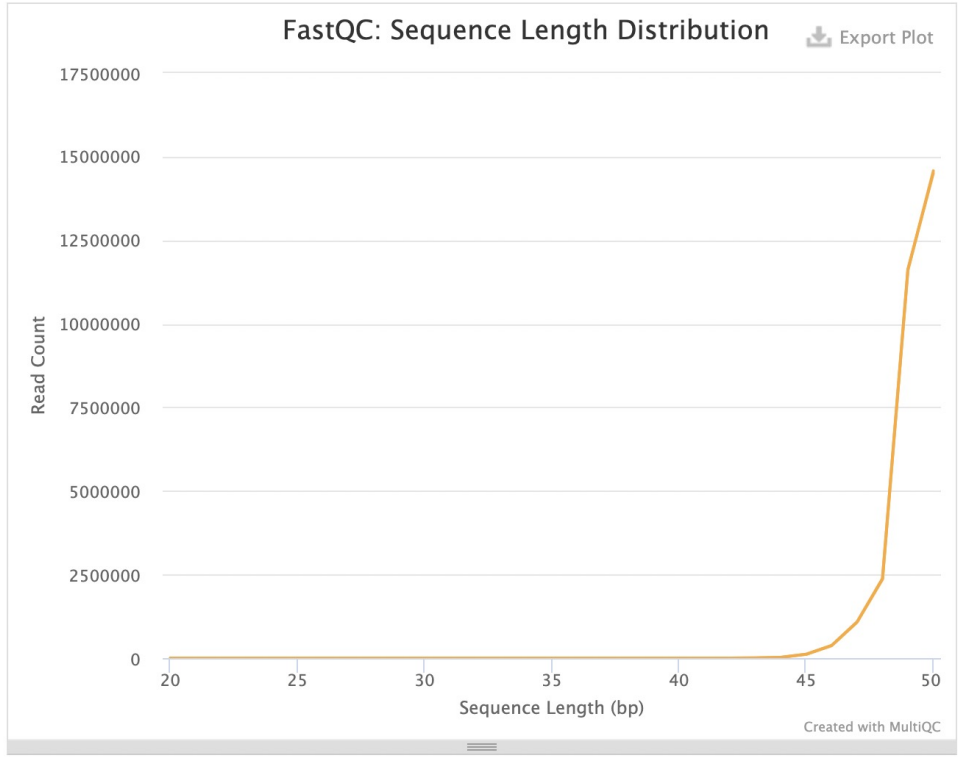
Sequence Length Distribution 80

All samples have sequences of a single length (50bp).



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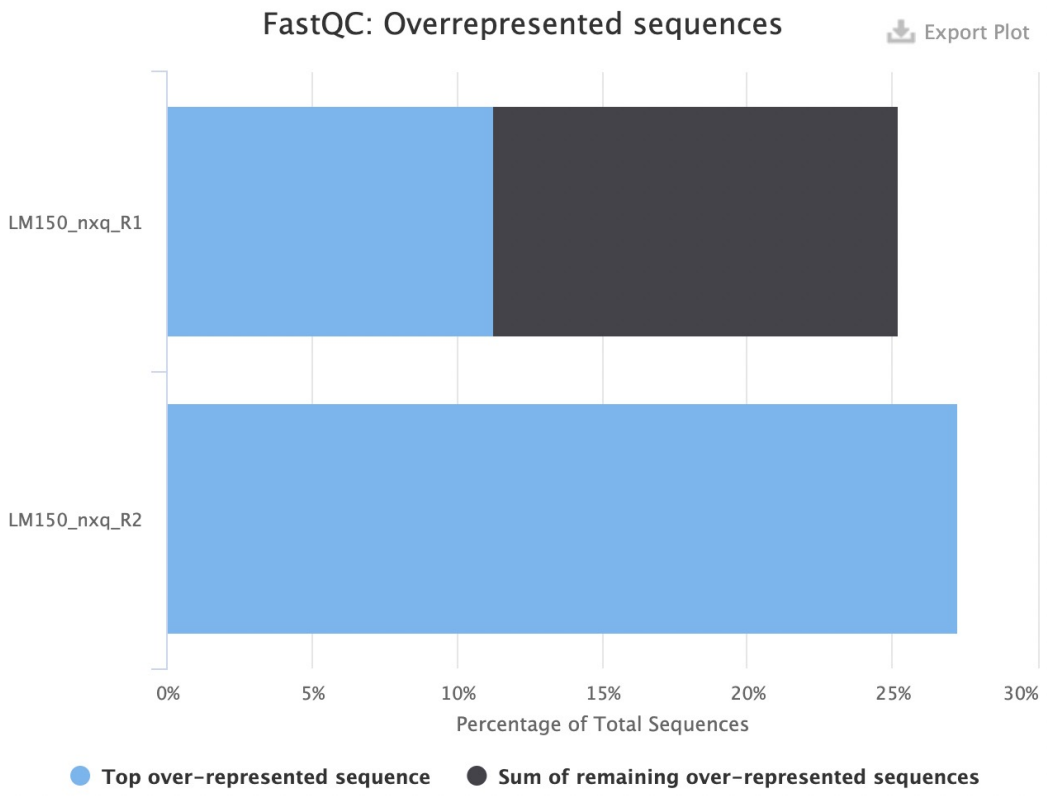

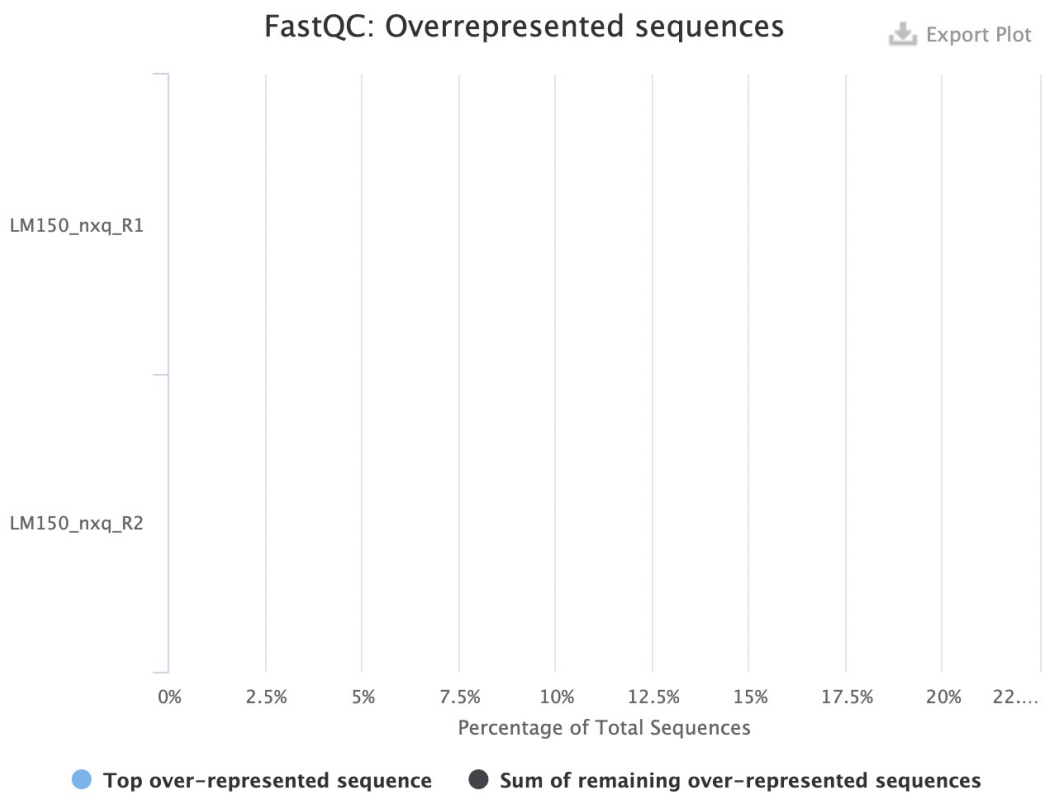

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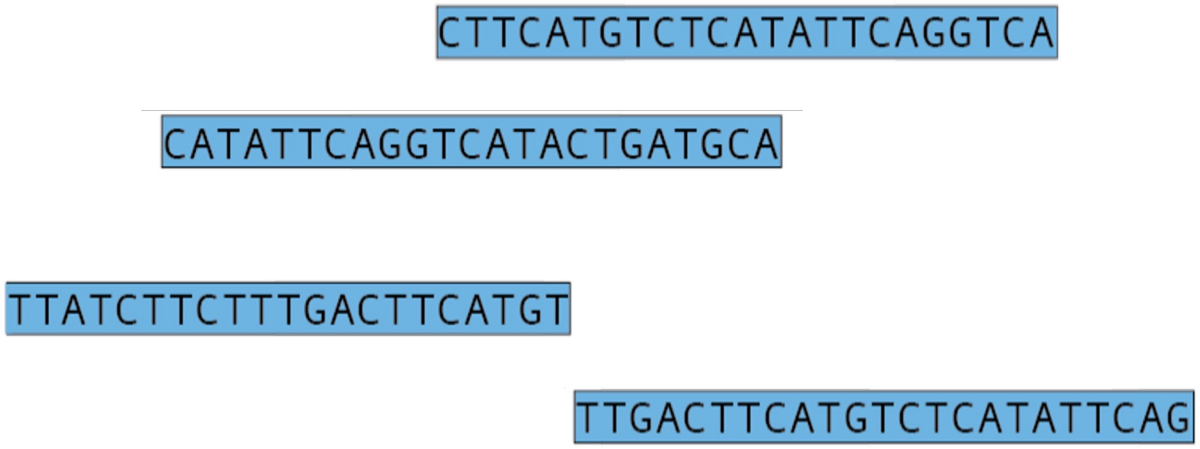
Mapping sequencing reads to a reference



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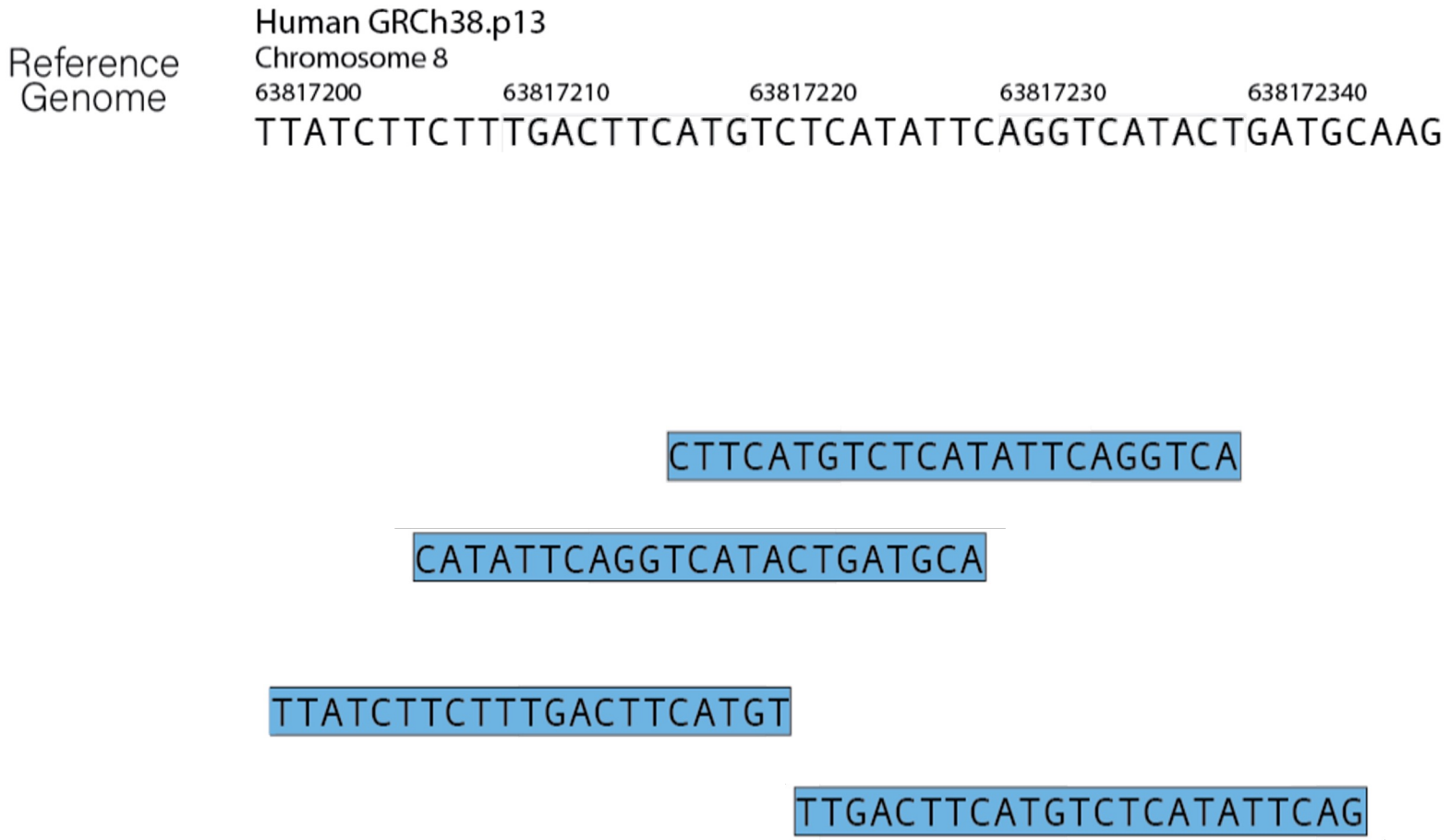

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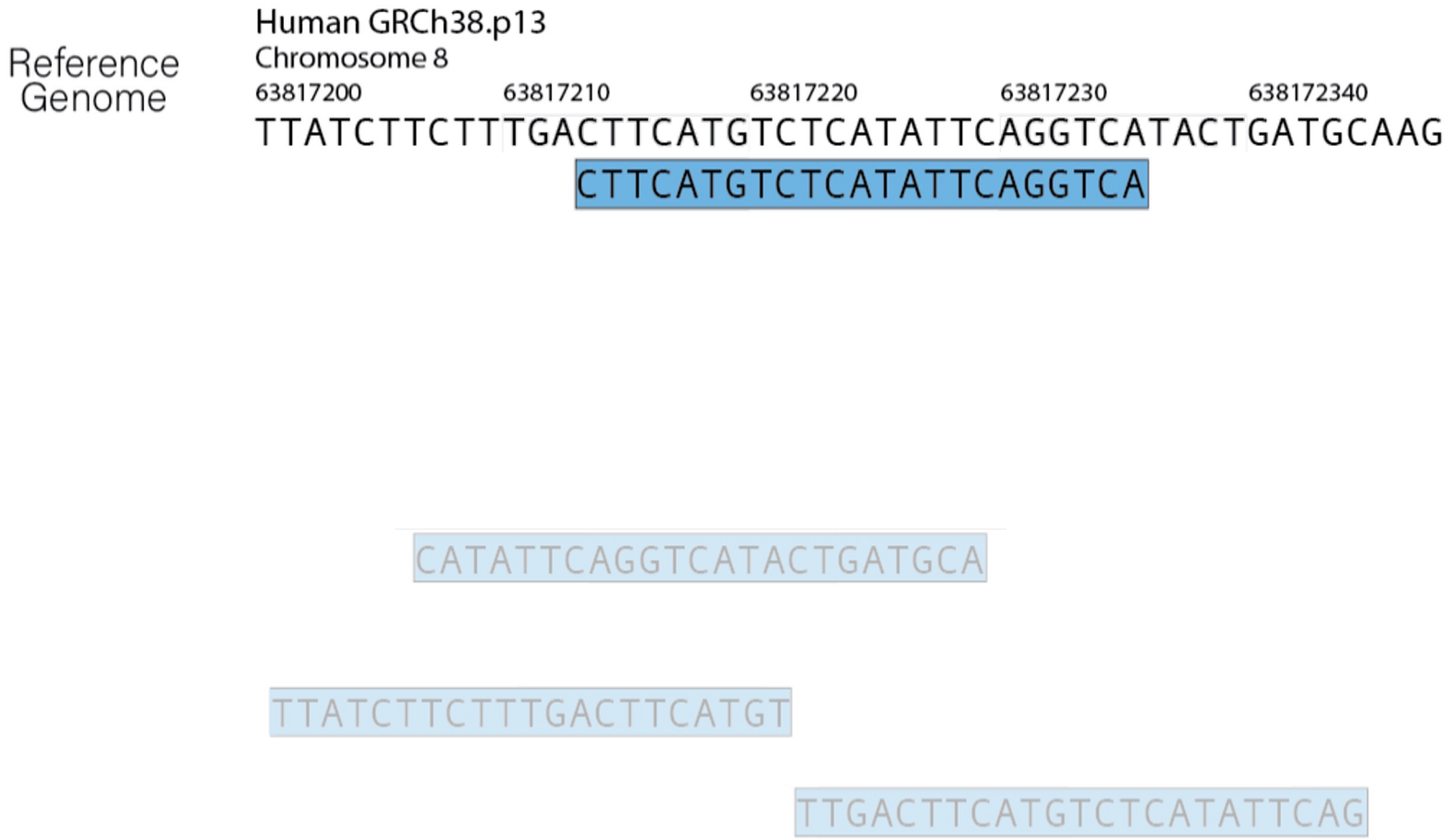

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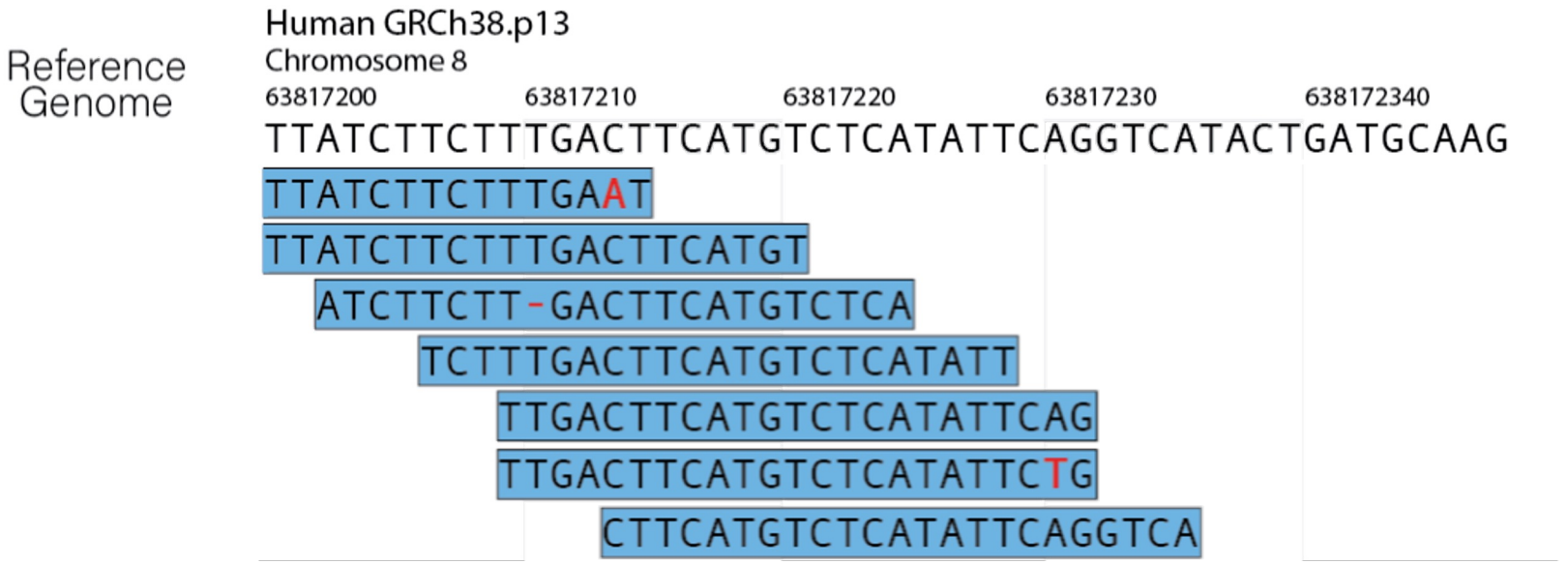

SAM file format

Sequence Alignment Map (SAM) is a human-readable, rectangular, text-based format for storing biological sequences aligned to a reference sequence.

Each entry (line) describes where a read is mapped on the reference and how it is mapped

Col	Field	Type	Brief description
1	QNAME	String	Query template NAME
2	FLAG	Int	bitwise FLAG
3	RNAME	String	References sequence NAME
4	POS	Int	1- based leftmost mapping POSition
5	MAPQ	Int	MAPping Quality
6	CIGAR	String	CIGAR string
7	RNEXT	String	Ref. name of the mate/next read
8	PNEXT	Int	Position of the mate/next read
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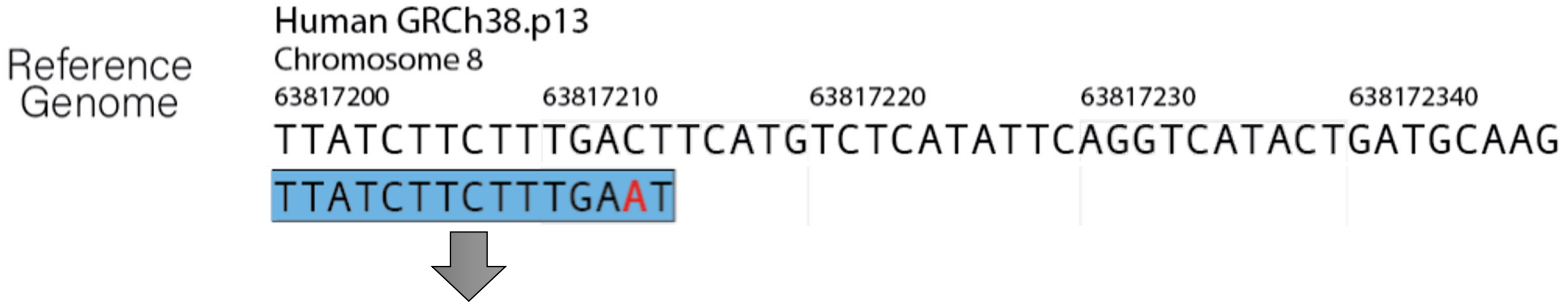


Read name	Flag	Chr	Position	Length	CIGAR	Read name (mate)	Chr (mate)	Position (mate)	Sequence	Per base sequencing quality
HWI-ST330:304:H045HADXX:2093#1	2	chr8	63817200	50	14M1X	2093#2	chr8	6381932	TTATCTTCTTTGAAT	?????BBBBBBDD=?

BAM file format

BAM files are **binarized** SAM files, allowing great compression of the alignment results.

However, bam files are not directly human-readable.

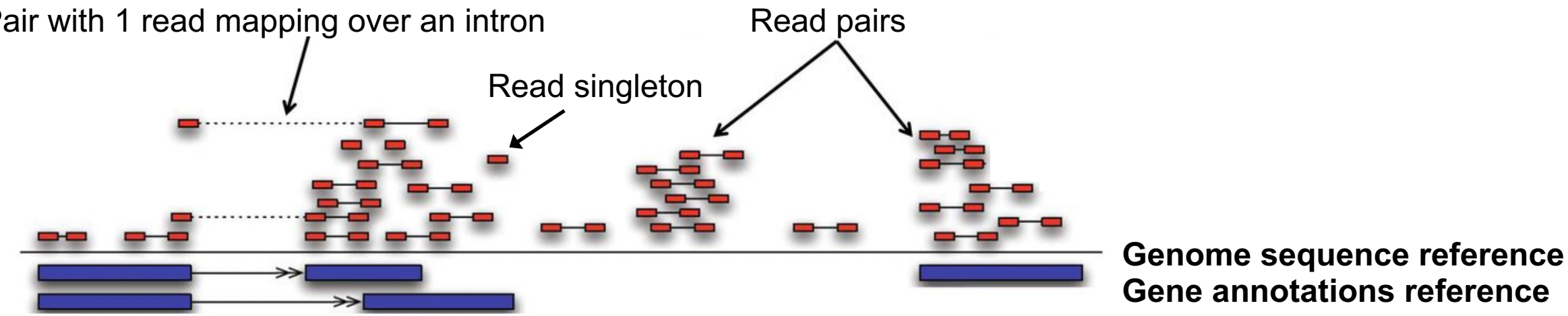
```
samtools view --bam ….sam > ….bam
```

Mapping tools

There are a plethora of alignment tools.

Each one requires the genome reference to be indexed first.

Some mappers can be ”**splice-aware**”, allowing reads to be mapped over annotated introns.



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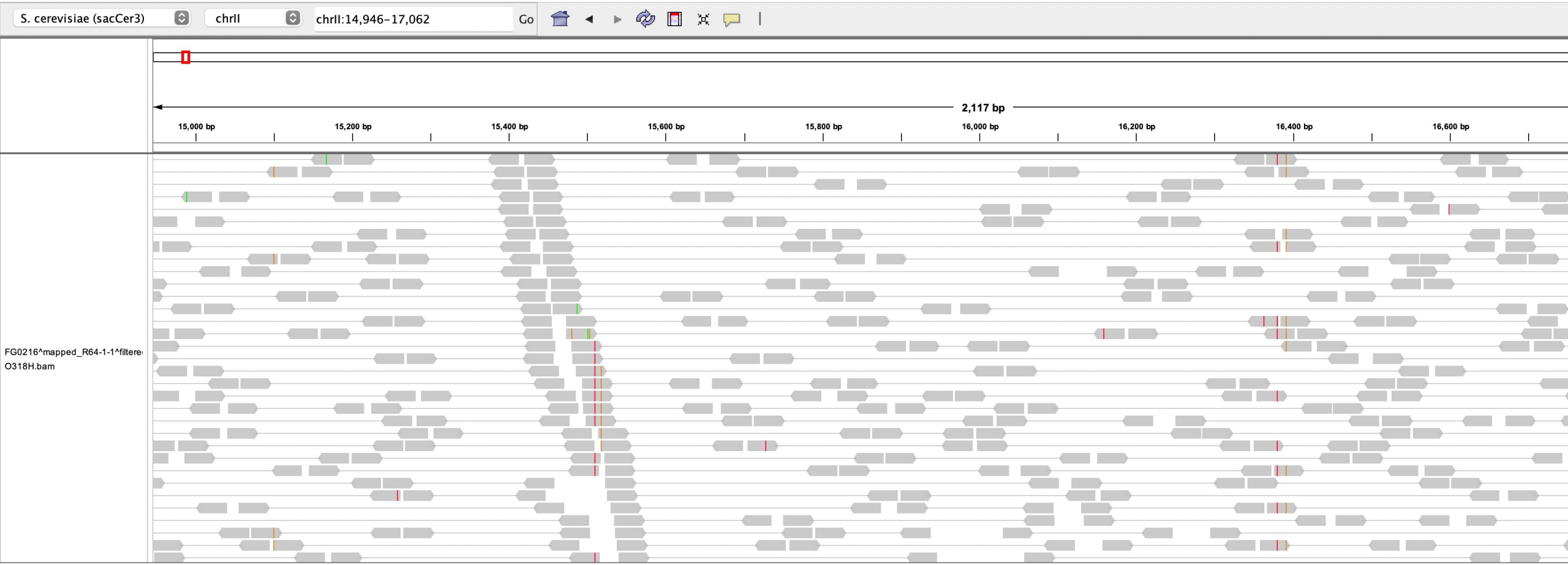
IGV: Integrative Genome Browser



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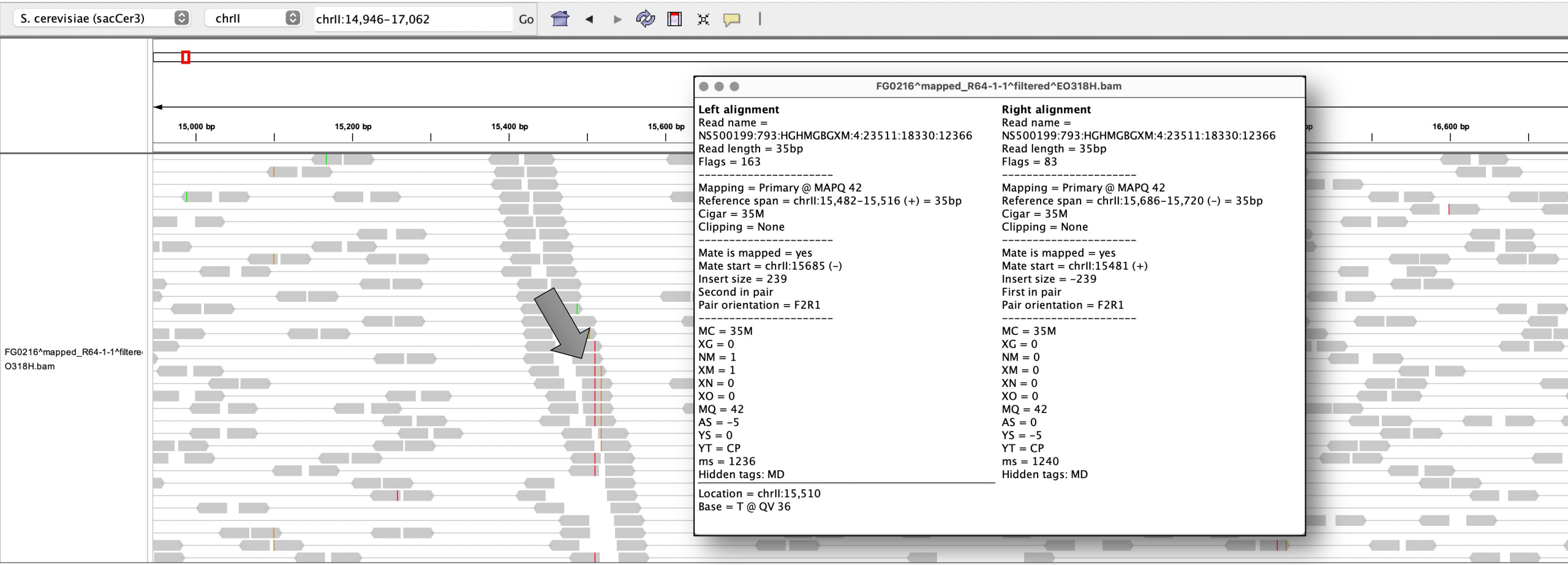

Filtering duplicates

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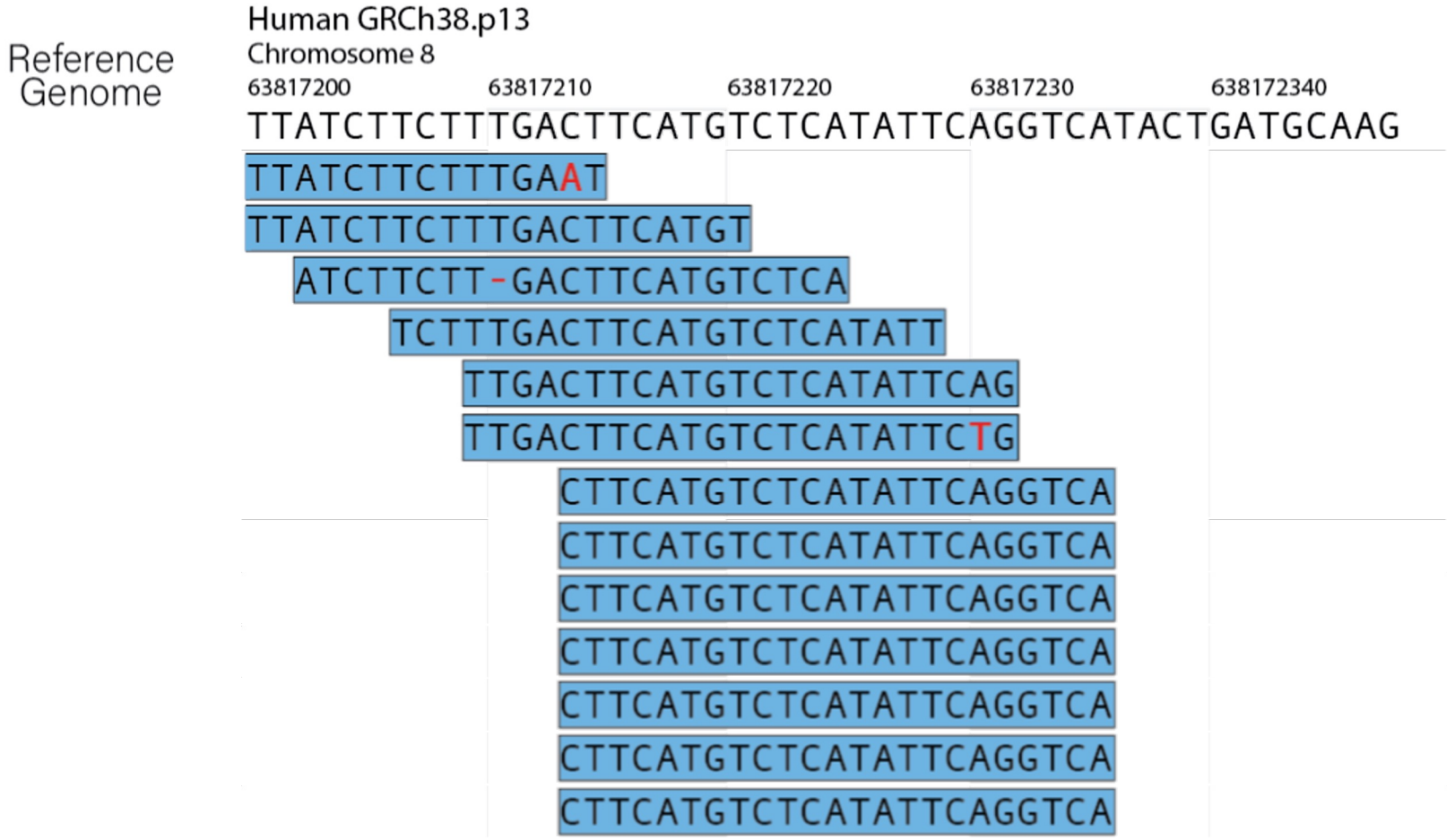

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Filtering out low-quality mapping reads

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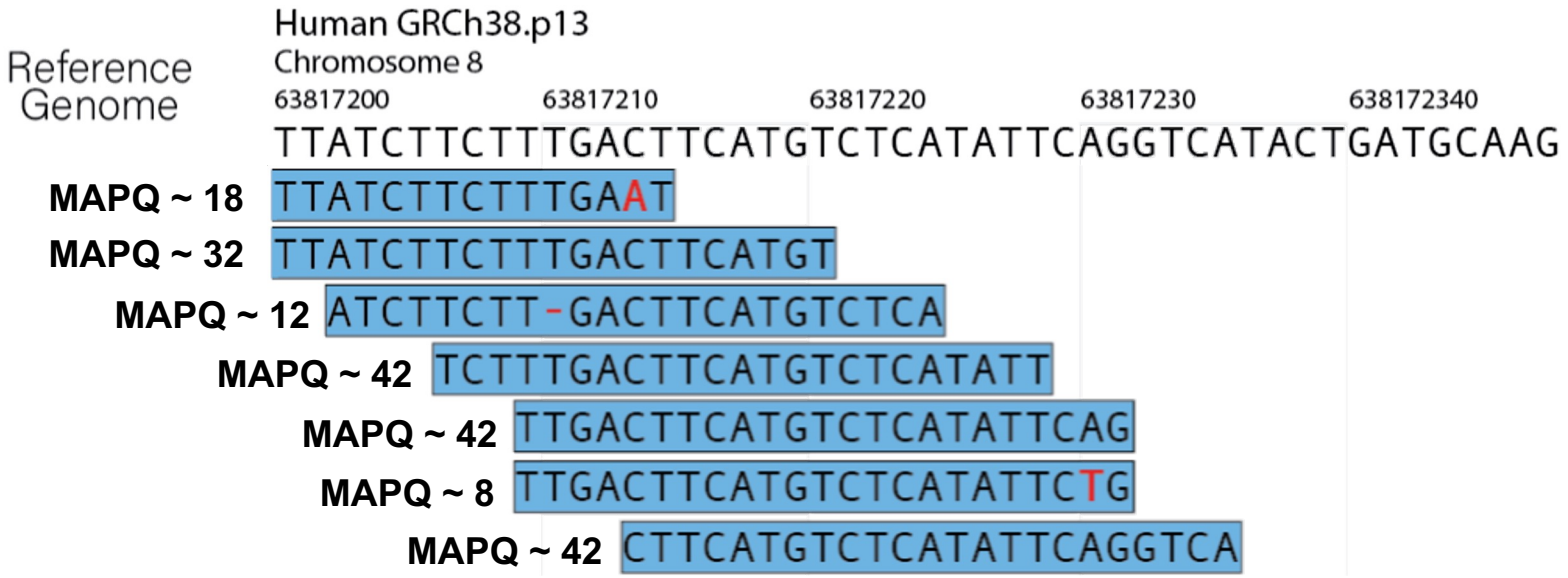
$$MAPQ = -10 * log_{10}\big(P(read\ is\ wrongly\ mapped)\big)$$



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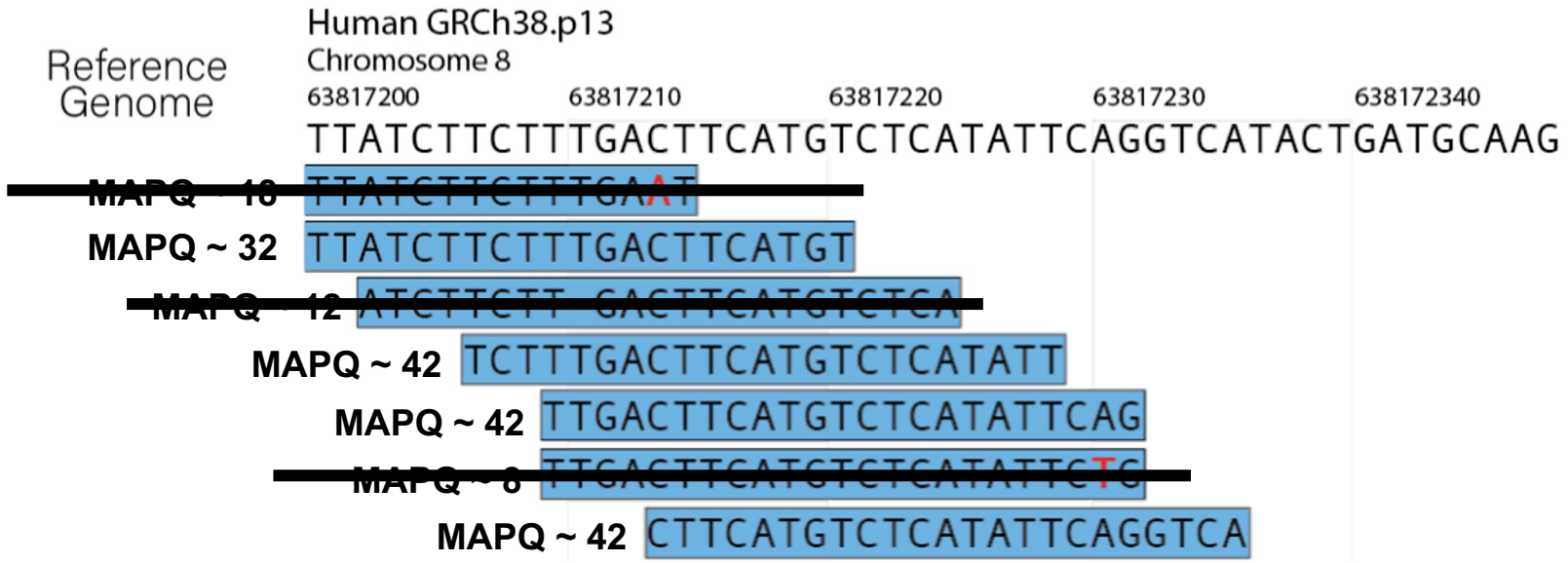
For example, a MAPQ score of 20 indicates that the probability for the read to be map at the indicated position is 0.01.



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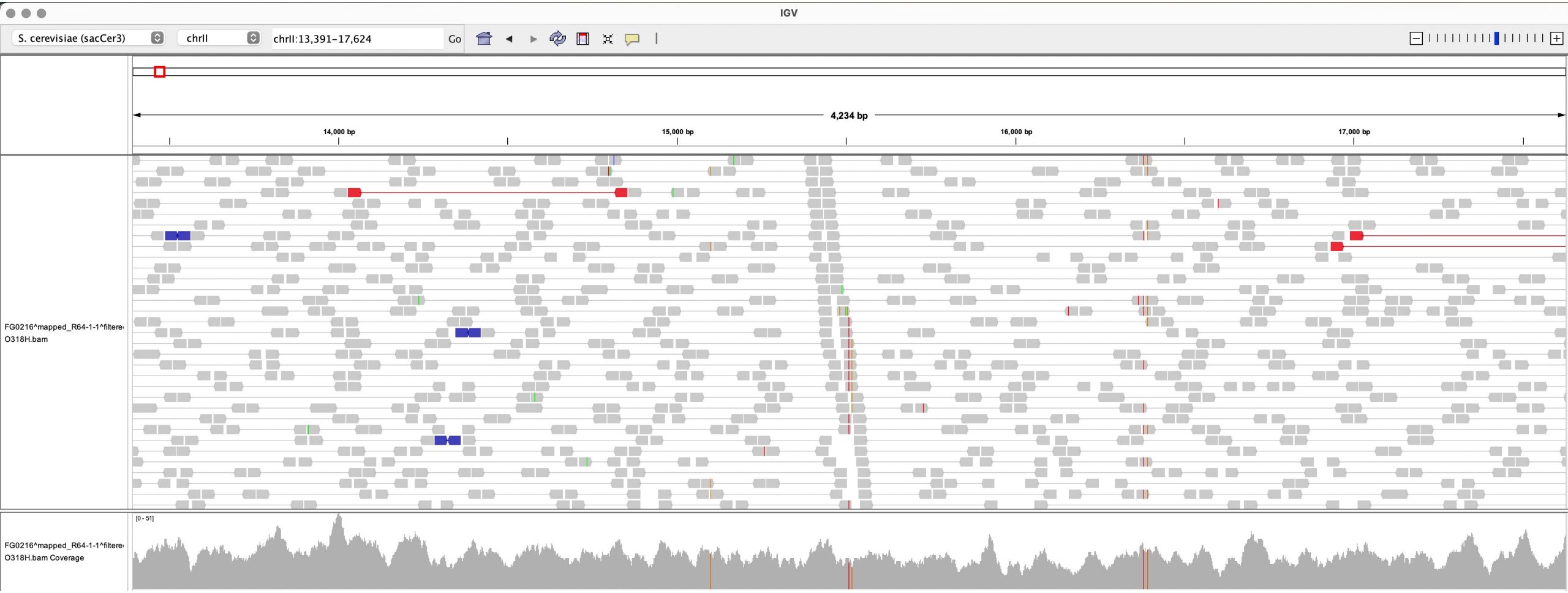
Generating tracks from mapped reads

Basic approach: ”pile-up” of all the fragments in `bam` files to generate a **coverage track**.



deepTools: a software suite to manage/produce genomic tracks

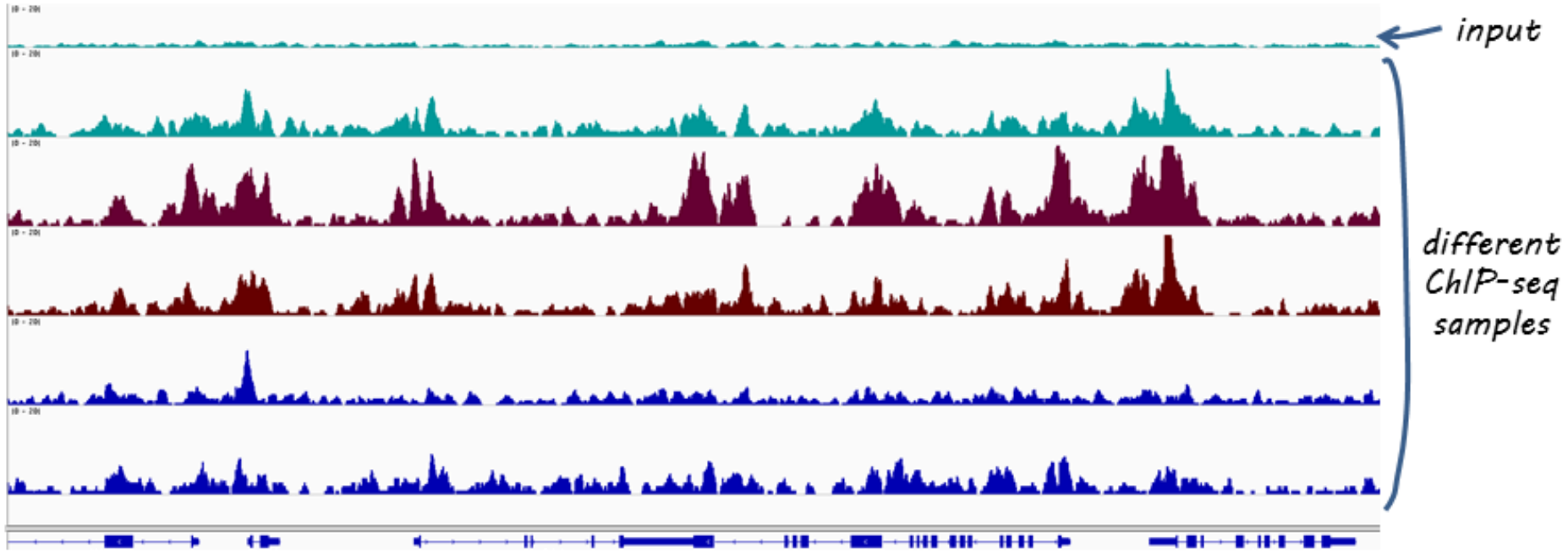
https://deeptools.readthedocs.io/en/latest/content/list_of_tools.html

- Tools for BAM and bigWig file processing
  - multiBamSummary
  - multiBigwigSummary
  - correctGCBias
  - bamCoverage
  - bamCompare
  - bigwigCompare
  - bigwigAverage
  - computeMatrix
  - alignmentSieve

- Tools for QC
  - plotCorrelation
  - plotPCA
  - plotFingerprint
  - bamPEFragmentSize
  - computeGCBias
  - plotCoverage
- Heatmaps and summary plots
  - plotHeatmap
  - plotProfile
  - plotEnrichment
- Miscellaneous
  - computeMatrixOperations
  - estimateReadFiltering

deepTools: a software suite to manage/produce genomic tracks

for visualizing continuous data, e·g· in the UCSC Genome Browser or IGV, bigWig files come in really handy



remember that there are 2 deepTools for bam → bigWig conversion:

- ***bamCoverage:*** *for individual files (like those shown here)*
- ***bamCompare:*** *to normalize two files to each other*

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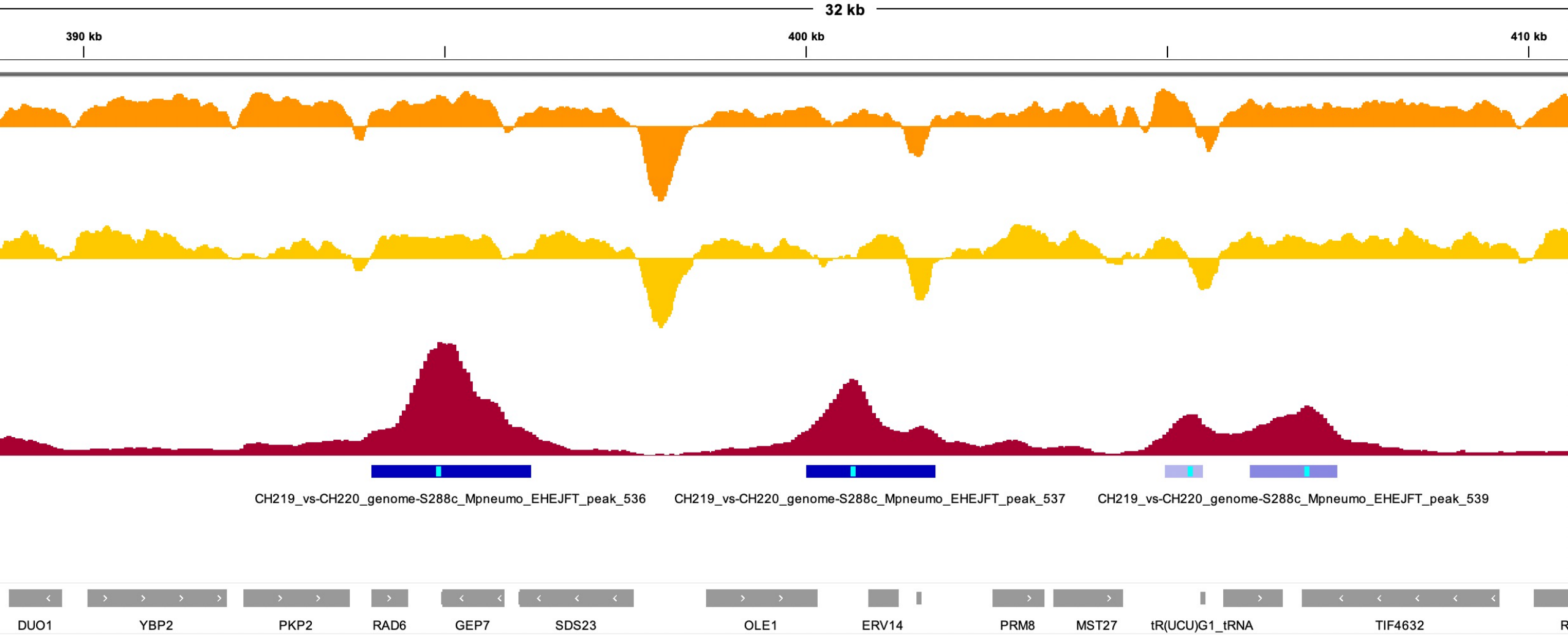
Epigenomics in a browser



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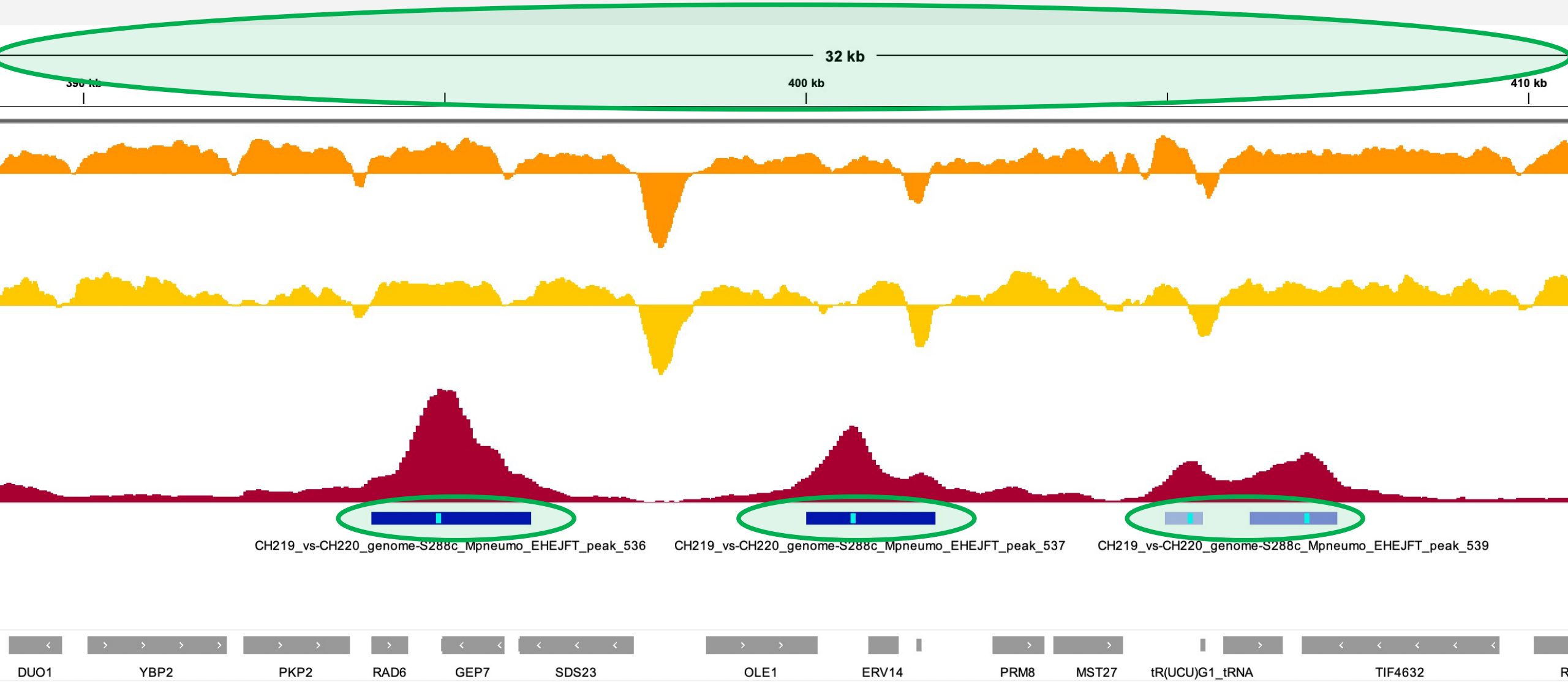
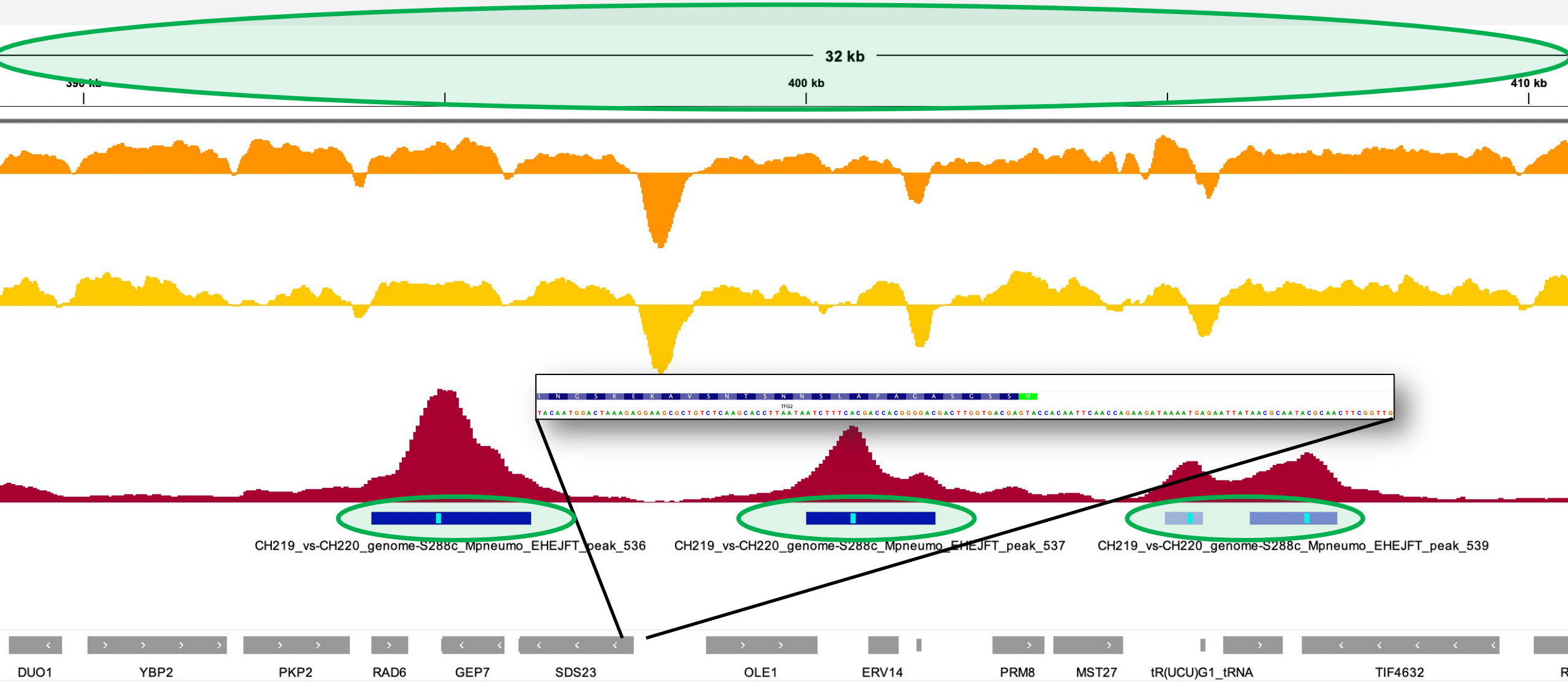


Epigenomics in a browser



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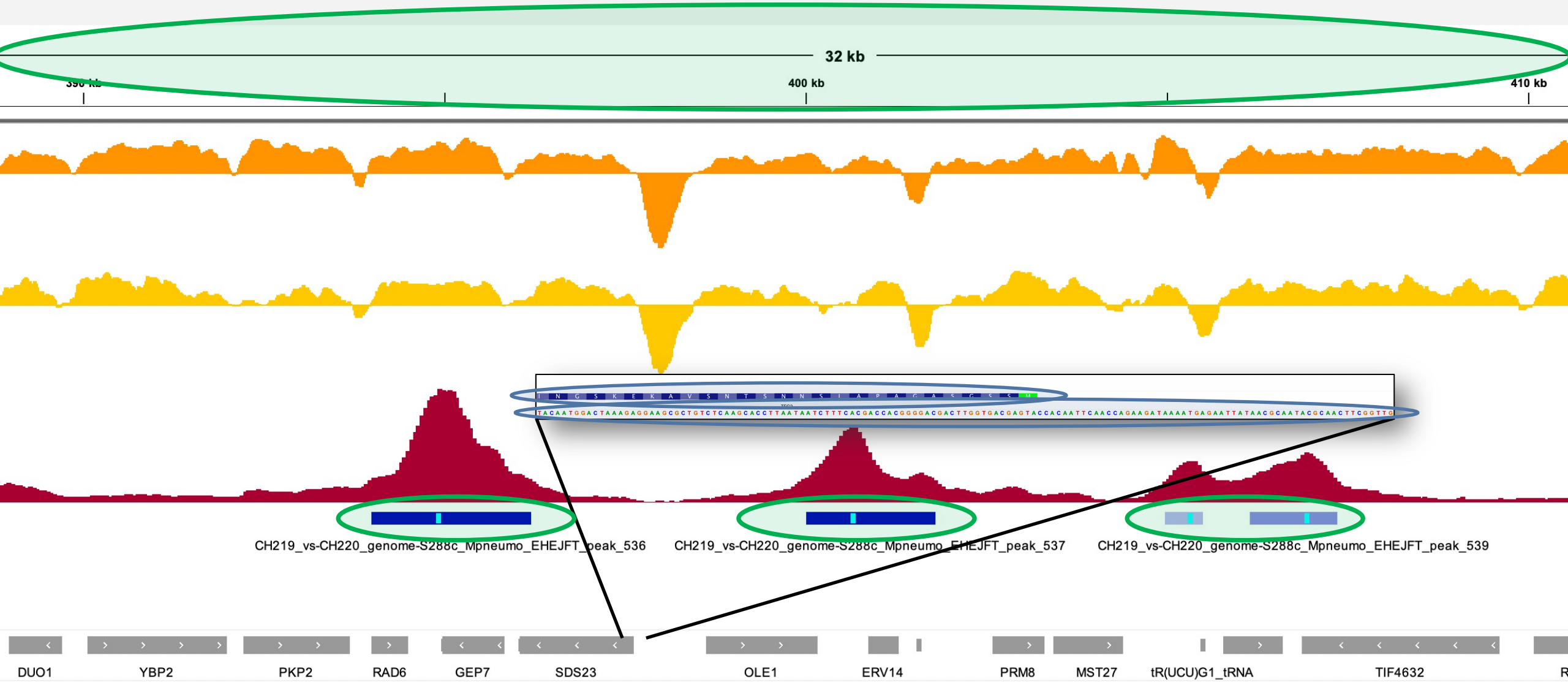

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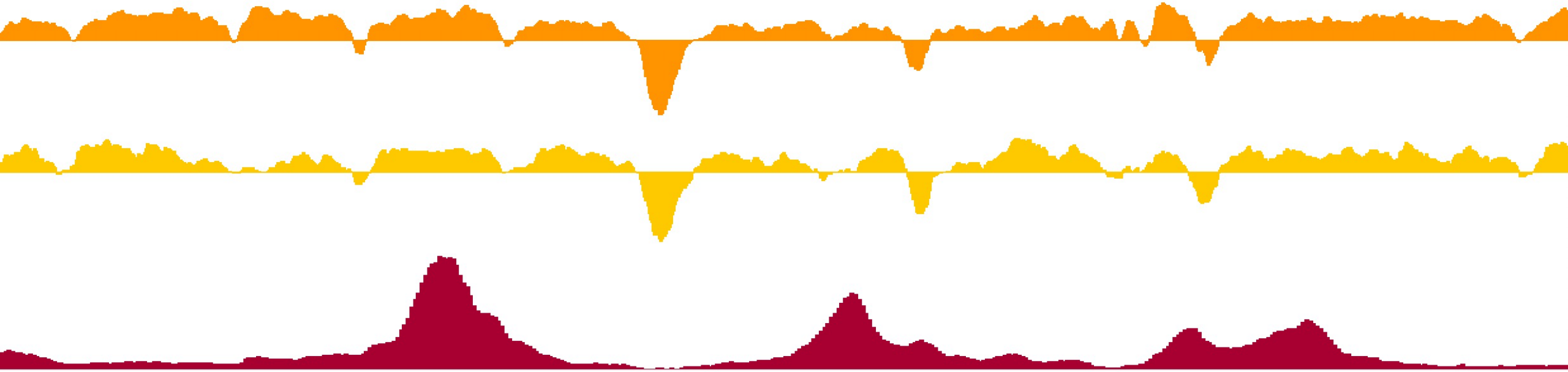

Epigenomics in a browser

- Genomic tracks are generally stored as bigwig files.
- bigwig files store long numerical vectors in a binarized format

```
I       2       5       0.153096
I       5       7       0.459288
I       7       9       0.612384
I       9       11      0.76548
I       11      15      0.918576
I       15      16      1.07167
I       16      17      1.37786
I       17      30      1.68406
```

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```
                                                
```

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- In R, bigwig files can be imported with `import()` from the `rtracklayer` package

```
> library(rtracklayer)

> import('….bw')

GRanges object with 6243328 ranges and 1 metadata column:
            seqnames    ranges strand |     score
               <Rle> <IRanges>  <Rle> | <numeric>
       [1]        I       3-5      * |  0.153096
       [2]        I       6-7      * |  0.459288
       [3]        I       8-9      * |  0.612384
       [4]        I     10-11      * |  0.765480
       [5]        I     12-15      * |  0.918576
       ...      ...       ...    ... .       ...
  [6243324]     Mito     85775      * |   9.79815
  [6243325]     Mito     85776      * |   8.26719
  [6243326]     Mito     85777      * |   6.43003
  [6243327]     Mito     85778      * |   5.66455
  [6243328]     Mito     85779      * |   3.21502
  -------
  seqinfo: 17 sequences from an unspecified genome
```

Run-length encoding vectors

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2 k

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```
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```
b b b b k k e e f a a a a a a a a g g g
```

```
Run-values:  b k e f a g
Run-lengths: 4 2 2 1 8 3
```

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```
b b b b k k e e f a a a a a a a a g g g
```

```
Run-values:  b k e f a g
Run-lengths: 4 2 2 1 8 3
```

12 alpha-numeric values instead of 20 alphabetic values

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```
> import('….bw', as = 'Rle')

RleList of length 17
$I
numeric-Rle of length 230218 with 104639 runs
  Lengths:         2         3         2         2         2 ...         2        30         1      1084
  Values :  0.000000  0.153096  0.459288  0.612384  0.765480 ...  0.612384  0.459288  0.306192  0.000000

$II
numeric-Rle of length 813184 with 424729 runs
  Lengths:         2         1         1         2         1 ...         6         1         5         2
  Values :  0.153096  0.306192  0.612384  0.918576  1.071670 ...  0.459288  0.306192  0.153096  0.000000

...
<15 more elements>
```